Industry Best Practices: Virtual Machines and Scale Sets

Use measured workload demand and business requirements to optimize compute safely. Cost reduction must not weaken availability, performance, security, recovery, or an approved capacity plan.

Applicable resource types

Microsoft.Compute/virtualMachines and Microsoft.Compute/virtualMachineScaleSets.

Cost optimization practices

Establish a baseline from CPU, memory, disk, network, queue depth, and workload-specific service-level indicators. Evaluate sustained peak demand, not only averages.

• Right-size only after observing representative business cycles, batch windows, and seasonal events.

• Use autoscale or schedules where workload patterns are predictable and the operational design supports them.

• Evaluate Spot capacity only for interruptible, fault-tolerant workloads with tested interruption handling.

• Consider reservations or savings plans only after removing avoidable waste and confirming stable, eligible demand.

Operational safeguards

A lower SKU or instance count is a controlled change, not a billing action. Preserve rollback capacity and validate application latency, error rate, and throughput during a pilot.

• Require workload-owner approval for production changes.

• Confirm high availability, zone, backup, licensing, and network throughput requirements before changing a size or family.

• Record the original configuration, success criteria, observation window, and rollback trigger.

Evidence and approval model

Decision area	Evidence to collect	Safe action gate
Rightsizing	Utilization percentiles, workload SLOs, peak/seasonal demand, SKU capabilities	Owner-approved pilot; compare service health before and after
Scheduling	Business-hours demand, maintenance window, operational dependencies	Approved schedule, start-up validation, exception process
Commitment discount	Optimized hourly baseline, eligibility, utilization forecast	Finance and owner review; avoid overlapping commitments

Source material

This reference summarizes the following official guidance. Check the linked sources for current service limits, billing terms, and implementation details.

- **Azure VM cost optimization best practices**
https://learn.microsoft.com/en-us/azure/virtual-machines/cost-optimization-best-practices

- **Azure Well-Architected VM guidance**
https://learn.microsoft.com/azure/well-architected/service-guides/virtual-machines

- **Azure reservations and savings plans**
https://learn.microsoft.com/azure/cost-management-billing/savings-plan/decide-between-savings-plan-reservation